

For second iteration:

- $w = 0.182$
- $b = 0.0645$
- $\alpha = 0.01$

$$\hat{Y} = 0.182 X + 0.0645$$

X	Y	$\hat{Y}$
1	2	0.2465
2	2.8	0.4285
3	3.6	0.6105
4	4.5	0.7925

- Gradient with respect to w :

$$\frac{\partial L}{\partial w} = \frac{2}{n} \sum_{i=1}^n (w \cdot X + b - y) \cdot X$$

$$\frac{\partial L}{\partial w} = \frac{2}{4} [(0.2465 - 2) \cdot 1 + (0.4285 - 2.8) \cdot 2 + (0.6105 - 3.6) \cdot 3 + (0.7925 - 4.5) \cdot 4]$$

$$\frac{\partial L}{\partial w} = -15.1475$$

$$\frac{\partial L}{\partial b} = \frac{2}{n} \sum_{i=1}^n (w \cdot X + b - y)$$

$$\frac{\partial L}{\partial b} = \frac{2}{4} [(0.2465 - 2) + (0.4285 - 2.8) + (0.6105 - 3.6) + (0.7925 - 4.5)]$$

$$\frac{\partial L}{\partial b} = -5.411$$

- Update parameters:

$$w = w - \alpha \cdot \frac{\partial L}{\partial w}$$

$$w = 0.182 - 0.01 * -15.1475$$

$$\mathbf{w = 0.333475}$$

$$b = b - \alpha \cdot \frac{\partial L}{\partial b}$$

$$b = 0.0645 - 0.01 * -5.411$$

$$\mathbf{b = 0.11861}$$